

Challenge 5 Solution

Math functions in C

Function	Meaning	Example	Result
sqrt(x)	Square root	sqrt(25)	5
pow(x,y)	x raised to y	pow(2,3)	8
ceil(x)	Round upward	ceil(4.2)	5
floor(x)	Round downward	floor(4.8)	4
round(x)	Nearest integer	round(4.6)	5
fabs(x)	Absolute value for floating point	fabs(-5.2)	5.2
fmod(x,y)	Floating-point remainder	fmod(10.5,3)	1.5
sin(x)	Sine	sin(x)	—
cos(x)	Cosine	cos(x)	—
tan(x)	Tangent	tan(x)	—
log(x)	Natural logarithm	log(10)	—
log10(x)	Base-10 logarithm	log10(100)	2
exp(x)	e ^x	exp(1)	≈ 2.718

```
#include <stdio.h>
#include <math.h>

int main()
{
    double n = 25;

    printf("Square root = %.2f\n", sqrt(n));
    printf("Power = %.2f\n", pow(2, 3));
    printf("Ceiling = %.2f\n", ceil(4.2));
    printf("Floor = %.2f\n", floor(4.8));
    printf("Absolute value = %.2f\n", fabs(-5.5));
}
```

```
return 0;

Square root = 5.00
Power = 8.00
Ceiling = 5.00
Floor = 4.00
Absolute value = 5.50
```

```
#include <stdio.h>
#include <math.h>

int main()
{
    int n = -25.5;

    printf("Absolute value = %.2f", fabs(n));

    return 0;
}
```

```
}  
Absolute value = 25.00  
#include <stdio.h>  
#include <stdlib.h>  
  
int main()  
{  
    int n = -25;  
  
    printf("Absolute value = %d", abs(n));  
  
    return 0;  
}
```

Absolute value = 25

```
#include <stdio.h>  
#include <math.h>  
  
int main()  
{  
    double a = 10.5, b = 3.0;  
  
    printf("Remainder = %.2f", fmod(a, b));  
  
    return 0;  
}
```

Remainder = 1.50

```
#include <stdio.h>  
#include <math.h>  
  
int main()  
{  
    double n = 4.2;  
  
    printf("Ceiling value = %.0f", ceil(n));  
  
    return 0;  
}
```

Ceiling value = 5

```
#include <stdio.h>  
#include <math.h>  
  
int main()  
{  
    double n = 4.8;  
  
    printf("Floor value = %0.0f", floor(n));  
  
    return 0;  
}
```

Floor value = 4

```
#include <stdio.h>
#include <math.h>

int main()
{
    double n = 4.4;

    printf("Rounded value = %0.0f", round(n));

    return 0;
}
```

Rounded value = 4

```
#include <stdio.h>
#include <math.h>

int main()
{
    double angle = 3.14159 / 2;

    printf("sin(90) = %.2f", sin(angle));

    return 0;
}
```

sin(90) = 1.00

```
#include <stdio.h>
#include <math.h>

int main()
{
    double angle = 0;

    printf("cos(0) = %.2f", cos(angle));

    return 0;
}
```

cos(0) = 1.00

```
#include <stdio.h>
#include <math.h>

int main()
{
    double n = 10;

    printf("Natural log = %.2f", log(n));

    return 0;
}
```

Natural log = 2.30

```
#include <stdio.h>
#include <math.h>
```

```
int main()
{
    double n = 8;

    printf("log with base 2 = %.2f", log2(n));

    return 0;
}
```

log with base 2 = 3.00

```
#include <stdio.h>
#include <math.h>

int main()
{
    double n = 100;

    printf("log with base 10 = %.2f", log10(n));

    return 0;
}
```

log with base 10 = 2.00

Challenge 5: Accepted



1. Square Root

Write a C program to read a positive number and find its square root.

Sample Input

Enter a number: 81

Sample Output

Square root = 9.00

```
#include <stdio.h>
#include <math.h>
int main() {
    float num, result;
    printf("Enter a number: ");
    scanf("%f", &num);
```

```
result = sqrt(num);
printf("Square root = %.2f\n", result);
return 0;
}
```

2. Power of a Number

Write a C program to read two numbers x and y and calculate x^y .

Sample Input

Enter x: 2

Enter y: 5

Sample Output

2 raised to the power 5 = 32.00

```
#include <stdio.h>
#include <math.h>

int main() {
    float x, y, result;
    printf("Enter x: ");
    scanf("%f", &x);
    printf("Enter y: ");
    scanf("%f", &y);
    result = pow(x, y);
    printf("%.0f raised to the power %.0f = %.2f\n", x, y, result);
    return 0;
}
```

3. Area of a Circle

Write a C program to read the radius of a circle and calculate its area.

$$Area = \pi r^2$$

Sample Input

Enter radius: 7

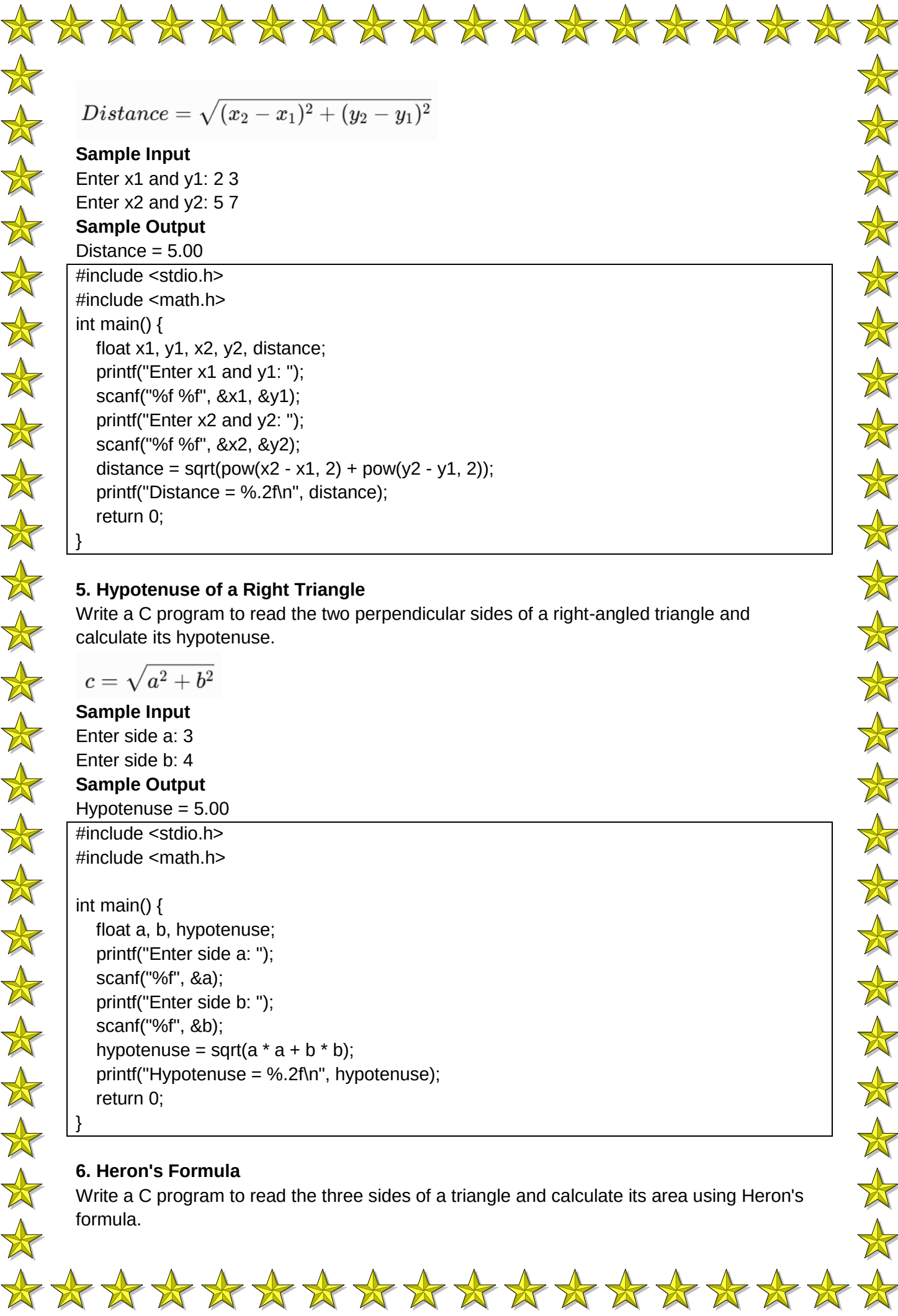
Sample Output

Area of circle = 153.94

```
#include <stdio.h>
#define PI=3.14159;
int main() {
    float radius, area;
    printf("Enter radius: ");
    scanf("%f", &radius);
    area = PI * radius * radius;
    printf("Area of circle = %.2f\n", area);
    return 0;
}
```

4. Distance between Two Points

Write a C program to read the coordinates (x1, y1) and (x2, y2) of two points and calculate the distance between them


$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Sample Input

Enter x1 and y1: 2 3

Enter x2 and y2: 5 7

Sample Output

Distance = 5.00

```
#include <stdio.h>
#include <math.h>
int main() {
    float x1, y1, x2, y2, distance;
    printf("Enter x1 and y1: ");
    scanf("%f %f", &x1, &y1);
    printf("Enter x2 and y2: ");
    scanf("%f %f", &x2, &y2);
    distance = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
    printf("Distance = %.2f\n", distance);
    return 0;
}
```

5. Hypotenuse of a Right Triangle

Write a C program to read the two perpendicular sides of a right-angled triangle and calculate its hypotenuse.

$$c = \sqrt{a^2 + b^2}$$

Sample Input

Enter side a: 3

Enter side b: 4

Sample Output

Hypotenuse = 5.00

```
#include <stdio.h>
#include <math.h>

int main() {
    float a, b, hypotenuse;
    printf("Enter side a: ");
    scanf("%f", &a);
    printf("Enter side b: ");
    scanf("%f", &b);
    hypotenuse = sqrt(a * a + b * b);
    printf("Hypotenuse = %.2f\n", hypotenuse);
    return 0;
}
```

6. Heron's Formula

Write a C program to read the three sides of a triangle and calculate its area using Heron's formula.

$$s = \frac{a + b + c}{2}$$

$$Area = \sqrt{s(s-a)(s-b)(s-c)}$$

Sample Input:

Enter three sides: 5 6 7

Sample Output:

Area = 14.70

```
#include <stdio.h>
#include <math.h>
int main() {
    float a, b, c, s, area;
    printf("Enter three sides: ");
    scanf("%f %f %f", &a, &b, &c);
    s = (a + b + c) / 2.0;
    area = sqrt(s * (s - a) * (s - b) * (s - c));
    printf("Area = %.2f\n", area);
    return 0;
}
```

7. Quadratic Equation

Write a C program to find the roots of the quadratic equation $ax^2 + bx + c = 0$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sample Input:

Enter a, b and c: 1 -5 6

Sample Output:

Root1 = 3.00

Root2 = 2.00

```
#include <stdio.h>
#include <math.h>

int main() {
    float a, b, c, discriminant, root1, root2;

    printf("Enter a, b and c: ");
    scanf("%f %f %f", &a, &b, &c);

    discriminant = b * b - 4 * a * c;

    root1 = (-b + sqrt(discriminant)) / (2 * a);
    root2 = (-b - sqrt(discriminant)) / (2 * a);

    printf("Root1 = %.2f\n", root1);
    printf("Root2 = %.2f\n", root2);
}
```



```
return 0;
```

```
}
```